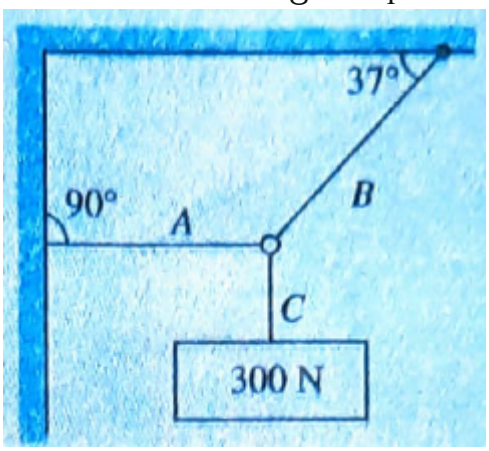
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A block of mass $30\ \text{kg}$ is suspended by three strings as shown in figure. Find the tension in string A.



(A) 175 N

(B) 185 N

(C) 200 N

(D) 400 N

Q.2

A string can withstand a tension of $25\ \text{N}$. What is the greatest speed at which a body of mass $1\ \text{kg}$ can be whirled in a horizontal circle using a $1\ \text{m}$ length of a string?

(A) $2.5\ \text{ms}^{-1}$

(B) $5.0\ \text{ms}^{-1}$

(C) $7.5\ \text{ms}^{-1}$

(D) $10\ \text{ms}^{-1}$

Q.3

A player stops a football weighing $0.5\ \text{kg}$ which comes flying towards him with a velocity of $10\ \text{m/s}$. If the impact lasts for $1/50$ th sec. and the ball bounces back with a velocity of $15\ \text{m/s}$, then the average force involved is

(A) $250\ \text{N}$

(B) $1250\ \text{N}$

(C) $500\ \text{N}$

(D) $625\ \text{N}$

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Q.4

A satellite in a force free space sweeps stationary interplanetary dust at a rate $(\mathrm{dM}/\mathrm{dt}) = \alpha \mathrm{v}$. The acceleration of satellite is

(A) $\frac{-2\alpha v^2}{M}$

(B) $\frac{-\alpha v^2}{\mathrm{M}}$

(C) $\frac{-\alpha v^2}{2\mathrm{M}}$

(D) $-\alpha v^2$

Q.5

A monkey is decending from the branch of a tree with constant acceleration. If the breaking strength is 75% of the weight of the monkey, the minimum acceleration with which monkey can slide down without breaking the branch is

(A) g

(B) $\frac{3\,\mathrm{g}}{4}$

(C) $\frac{\mathrm{g}}{4}$

(D) $\frac{\mathrm{g}}{2}$

Q.6

A stationary body of mass $3\ \mathrm{kg}$ explodes into three equal pieces. Two of the pieces fly off in two mutually perpendicular directions, one with a velocity of $3\hat{\mathrm{i}}\mathrm{ms}^{-1}$ and the other with a velocity of $4\hat{\mathrm{j}}\mathrm{ms}^{-1}$. If the explosion occurs in $10^{-4}\ \mathrm{s}$, the average force acting on the third piece in newton is

(A) $(3\hat{\mathrm{i}} + 4\hat{\mathrm{j}}) \times 10^{-4}$

(B) $(3\hat{\mathrm{i}} - 4\hat{\mathrm{j}}) \times 10^{-4}$

(C) $(3\hat{\mathrm{i}} - 4\hat{\mathrm{j}}) \times 10^{4}$

(D) $-(3\hat{\mathrm{i}} + 4\hat{\mathrm{j}}) \times 10^{4}$

Q.7

A bullet is fired from a gun. The force on the bullet is given by $F = 600 - 2 \times 10^5 t$ where, F is in newton and t in second. The force on the bullet becomes zero as soon as it leaves the barrel. What is the average impulse imparted to the bullet?

(A) $1.8\ \mathrm{N} - \mathrm{s}$

(B) zero

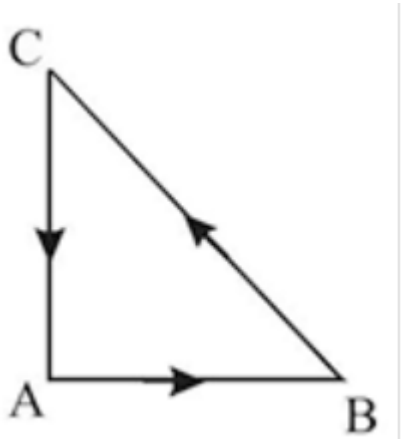
(C) $9\ \mathrm{N} - \mathrm{s}$

(D) $0.9\ \mathrm{N} - \mathrm{s}$

Q.8

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Three forces start acting simultaneously on a particle moving with velocity, $\vec{v}$. These forces are represented in magnitude and direction by the three sides of a triangle ABC. The particle will now move with velocity



(A) less than $\vec{v}$

(B) greater than $\vec{v}$

(C) $|\mathrm{v}|$ in the direction of the largest force BC

(D) $\vec{v}$, remaining unchanged

Q.9

A ball of mass $10\ \mathrm{g}$ moving perpendicular to the plane of the wall strikes it and rebounds in the same line with the same velocity. If the impulse experienced by the wall is $0.54\mathrm{Ns}$, the velocity of the ball is

(A) $27\ \mathrm{ms}^{-1}$

(B) $3.7\ \mathrm{ms}^{-1}$

(C) $54\ \mathrm{ms}^{-1}$

(D) $37\ \mathrm{ms}^{-1}$

Q.10

A car having a mass of $1000\ \mathrm{kg}$ is moving at a speed of 30 metres/sec. Brakes are applied to bring the car to rest. If the frictional force between the tyres and the road surface is 5000 newtons, the car will come to rest in

(A) 5 seconds

(B) 10 seconds

(C) 12 seconds

(D) 6 seconds

Q.11

A body of mass $5\ \mathrm{kg}$ under the action of constant force $\overrightarrow{\mathrm{F}} = \mathrm{F_x}\hat{\mathrm{i}} + \mathrm{F_y}\hat{\mathrm{j}}$ has velocity at $\mathrm{t} = 0\ \mathrm{s}$ as $\overrightarrow{\mathrm{v}} = (6\hat{\mathrm{i}} - 2\hat{\mathrm{j}})\mathrm{m/s}$ and at $\mathrm{t} = 10\ \mathrm{s}$ as $\overrightarrow{\mathrm{v}} = +6\hat{\mathrm{j}}\mathrm{m/s}$

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The force F is:

(A) $(-3\hat{\mathrm{i}}+4\hat{\mathrm{j}})\mathrm{N}$

(B) $\left(-\frac{3}{5}\hat{\mathrm{i}}+\frac{4}{5}\hat{\mathrm{j}}\right)\mathrm{N}$

(C) $(3\hat{\mathrm{i}}-4\hat{\mathrm{j}})\mathrm{N}$

(D) $\left(\frac{3}{5}\hat{\mathrm{i}}-\frac{4}{5}\hat{\mathrm{j}}\right)\mathrm{N}$

Q.12

A $5000\ \mathrm{kg}$ rocket is set for vertical firing. The exhaust speed is $800\ \mathrm{m/s}$. To give an initial upward acceleration of $20\ \mathrm{m/s^2}$, the amount of gas ejected per second (in $\mathrm{kg/sec}$) to supply the needed thrust will be (Take $\mathrm{g}=10\ \mathrm{m/s^2}$)

(A) 187.6

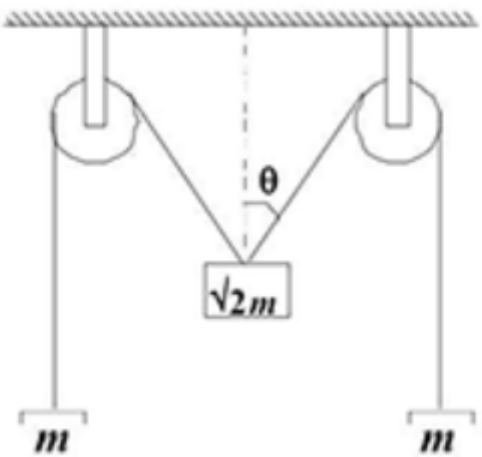
(B) 188.6

(C) 188.5

(D) 187.5

Q.13

The pulleys and strings shown in the figure are smooth and of negligible mass. For the system to remain in equilibrium, the angle θ (in degree) should be



(A) 44

(B) 45

(C) 46

(D) 47

Q.14

A pail filled with sand has a total mass of $60\ \mathrm{kg}$. A crane is lowering it such that it has an initial downward acceleration of $1.5\ \mathrm{m/s^2}$. A hole in the pail allows sand to leak out. If the force exerted by the crane on the pail does not change, what mass of sand must leak out before the downward acceleration decreases to zero?

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(A) $9.2\ \text{kg}$

(B) $20\ \text{kg}$

(C) $40\ \text{kg}$

(D) $51\ \text{kg}$

Q.15

A particle tied to a string describes a vertical circular motion of radius r continually. If it has a velocity $\sqrt{3\mathrm{gr}}$ at the highest point, then value of ratio of the respective tensions in the string holding it at the highest and lowest points is

(A) 0.23

(B) 0.24

(C) 0.21

(D) 0.25

Q.16

A mass of $1\ \text{kg}$ is suspended by a thread. It is
(i) lifted up with an acceleration $4.9\ \text{m/s}^2$,
(ii) lowered with an acceleration $4.9\ \text{m/s}^2$. The ratio of the tensions is

(A) $3:1$

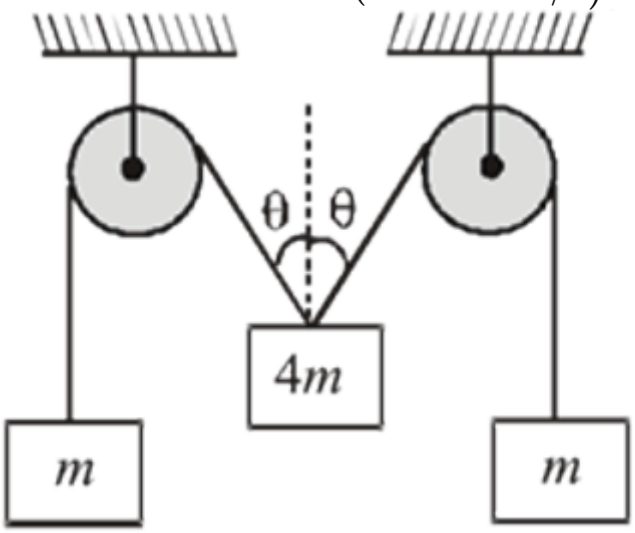
(B) $1:2$

(C) $1:3$

(D) $2:1$

Q.17

In the figure shown, the pulleys and strings are massless. The acceleration of the block of mass $4m$ just after the system is released from rest is $(\theta = \sin^{-1} 3/5)$



(A) $\frac{2g}{5}$ downwards

(B) $\frac{2g}{5}$ upwards

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(C) $\frac{5g}{11}$ downwards

(D) $\frac{5}{11}$ upwards

Q.18

Three blocks A, B and C of masses $4\ \text{kg}, 2\ \text{kg}$ and 1 kg respectively, are in contact on a frictionless surface, as shown. If a force of $14\ \text{N}$ is applied on the $4\ \text{kg}$ block then the contact force between A and B is



(A) $6\ \text{N}$

(B) 8N

(C) $18\ \text{N}$

(D) 2N

Q.19

A balloon with mass ' m ' is descending down with an acceleration ' a ' (where $a < g$). How much mass should be removed from it so that it starts moving up with an acceleration 'a'?

(A) $\frac{2\text{ma}}{\text{g}+\text{a}}$

(B) $\frac{2ma}{g-a}$

(C) $\frac{\text{ma}}{\text{g}+\text{a}}$

(D) $\frac{\text{ma}}{\text{g}-\text{a}}$

Q.20

A uniform hollow cylinder having mass m and radius r has a massless string wound over it. One end of string is attached to a mass m. If during the motion the string does not slip on the cylinder, and the mass falls with an acceleration a in cm / s^2 on release then give your answer as $\frac{a}{100}$. (Take $\text{g} = 9.8 \quad \text{m} / \text{s}^2$)

(A) 04.95

(B) 04.85

(C) 04.80

(D) 04.90

Answers & Solutions

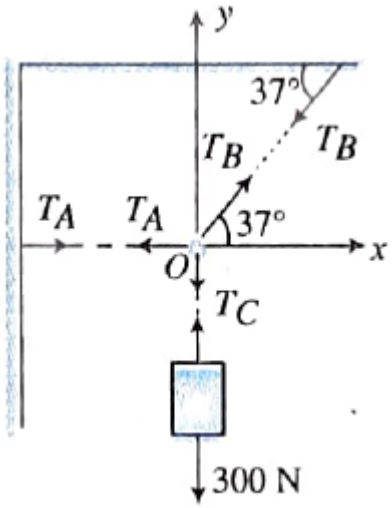
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Q.1 Answer:

400 N

Solution:

Method I: Considering equilibrium of each part of system The whole system is in equilibrium; therefore, for each part $\sum \vec{F} = 0$. From the free-body diagram of block $C, T_C = 300$ N.



Now consider the equilibrium of point O,

$\sum F_x = 0$ or $T_B \cos 37^\circ - T_A = 0$

$\therefore \quad T_A = T_B \cos 37^\circ = T_B \cdot \frac{4}{5}$ **...(i)**

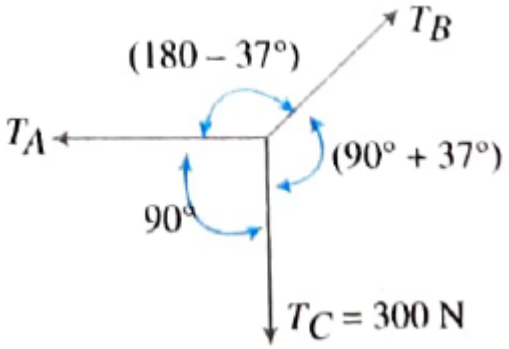
and $\sum F_y = 0$ **or** $T_B \sin 37^\circ - T_C = 0$ **...(ii)**

$\therefore \quad T_B = \frac{T_C}{\sin 37^\circ} = \frac{300}{3/5} = 500$ N

From Eq. (i), we get

$T_A = \frac{4}{5} T_B = \frac{4}{5} \times 500 = 400$ N

Method II: Using Lami's theorem



By Lami's theorem, we have

$\frac{T_A}{\sin(90+37^\circ)} = \frac{T_B}{\sin 90^\circ} = \frac{T_C}{\sin(180-37^\circ)}$

But $\quad TC = 300$ N

and $\quad T_B = \frac{T_C}{\sin 37^\circ} = \frac{300}{3/5} = 500$ N

$T_A = T_C \left(\frac{\sin(90+37^\circ)}{\sin(180-37^\circ)}\right) = 300 \left(\frac{\cos 37^\circ}{\sin 37^\circ}\right) = 400$ N

Q.2 Answer:

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$5.0\ \mathrm{ms}^{-1}$

Solution:

Since the body is whirled in a horizontal circle, the gravity, acting vertically downwards, has no effect on the motion. If v is the greatest speed with which the body can be whirled, the maximum centripetal force (or tension) in the string is mv^2/R, which must balance a force of $25\ \mathrm{N}$. Thus

$25 = \frac{mv^2}{R} = \frac{1 \times v^2}{1}$

which gives $v = 5\ \mathrm{ms}^{-1}$, which is choice (b).

Q.3 Answer:

$625\ \mathrm{N}$

Solution:

$$\begin{aligned} &\text{Here } m = 0.5\ \mathrm{kg}; \mathrm{u} = -10\ \mathrm{m/s} \\ &\mathrm{t} = 1/50\ \mathrm{s}; \mathrm{v} = +15\ \mathrm{ms}^{-1} \\ &\text{Force } = \mathrm{m(v-u)/t} = 0.5(10+15) \times 50 = 625\ \mathrm{N} \end{aligned}$$

Q.4 Answer:

$\frac{-\alpha v^2}{\mathrm{M}}$

Solution:

Thrust on the satellite,

$\mathrm{F} = \frac{-\mathrm{vdM}}{\mathrm{dt}} = -\mathrm{v}(\alpha \mathrm{v}) = -\alpha \mathrm{v}^2$

Acceleration $= \frac{\mathrm{F}}{\mathrm{M}} = \frac{-\alpha \mathrm{v}^2}{\mathrm{M}}$

Q.5 Answer:

$\frac{\mathrm{g}}{4}$

Solution:

Let T be the tension in the branch of a tree when monkey is descending with acceleration a. Then $\mathrm{mg} - \mathrm{T} = \mathrm{ma}$; **and** $\mathrm{T} = 75\%$ **of weight of monkey**

$$\begin{aligned} &= \mathrm{mg} - (75/100)\mathrm{mg} = \mathrm{ma} \\ &\Rightarrow (1/4)\mathrm{mg} = \mathrm{ma} \\ &\Rightarrow a = \tfrac{\mathrm{g}}{4} \end{aligned}$$

Q.6 Answer:

$-(3\hat{\mathrm{i}} + 4\hat{\mathrm{j}}) \times 10^4$

Solution:

According to law of conservation of momentum the third piece has momentum $= 1 \times -(3\hat{\mathrm{i}} + 4\hat{\mathrm{j}})\mathrm{kgms}^{-1}$

Impulse = Average force × time ⇒ Average force = $\frac{\text{Impulse}}{\text{time}} = \frac{\text{Change in momentum}}{\text{time}} = \frac{-(3\hat{\mathrm{i}}+4\hat{\mathrm{j}})\mathrm{kgms}^{-1}}{10^{-4}\ \mathrm{s}}$

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Q.7 Answer:

$0.9\ \mathrm{N}-\mathrm{s}$

Solution:

Given $\mathrm{F}=600-\left(2 \times 10^{5} \mathrm{t}\right)$

The force is zero at time t**, given by**

$$0=600-2 \times 10^{5} \mathrm{t}$$

$$\Rightarrow \quad t=\frac{600}{2 \times 10^{5}}=3 \times 10^{-3} \text{ seconds}$$

$$\therefore \text{ Impulse } =\int_{0}^{t} F dt=\int_{0}^{3 \times 10^{-3}}\left(600-2 \times 10^{5} t\right) dt$$

$$=\left[600 t-\frac{2 \times 10^{5} t^{2}}{2}\right]_{0}^{3 \times 10^{-3}}$$

$$=600 \times 3 \times 10^{-3}-10^{5}\left(3 \times 10^{-3}\right)^{2}$$

$$=1.8-0.9=0.9 \mathrm{Ns}$$

Q.8 Answer:

$\vec{v}$, remaining unchanged

Solution:

As shown in the figure, the three forces are represented by the sides of a triangle taken in the same order. Therefore the resultant force is zero. $\vec{F}_{net}=m\vec{a}$**. Therefore acceleration is also zero i.e., velocity remains unchanged.**

Q.9 Answer:

$27\ \mathrm{ms}^{-1}$

Solution:

As the ball, $\mathrm{m}=10\ \mathrm{g}=0.01\ \mathrm{kg}$ **rebounds after striking the wall** $\therefore$ **Change in momentum** $=mv-(-mv)=2mv$

Inpulse = Change in momentum $=2mv$

$\therefore v=\frac{\text{Impulse}}{2m}=\frac{0.54 \mathrm{Ns}}{2 \times 0.01\ \mathrm{kg}}=27\ \mathrm{m\ s}^{-1}$

Q.10 Answer:

6 seconds

Solution:

$$\mathrm{v}=\mathrm{u}-\mathrm{at} \Rightarrow \mathrm{t}=\frac{\mathrm{u}}{\mathrm{a}}[\mathrm{Asv}=0]$$

$$\mathrm{t}=\frac{\mathrm{u} \times \mathrm{m}}{\mathrm{F}}=\frac{30 \times 1000}{5000}=6 \mathrm{sec}$$

Q.11 Answer:

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$(-3\hat{\mathrm{i}}+4\hat{\mathrm{j}})\mathrm{N}$

Solution:

From question, Mass of body, $m = 5\ \mathrm{kg}$ **Velocity at** $t = 0, u = (6\hat{i} - 2\hat{j})\mathrm{m/s}$
Velocity at $t = 10\ \mathrm{s}, v = +6\hat{j}\ \mathrm{m/s}$
Force, $F = ?\ a = \frac{v-u}{t} = \frac{6\hat{j}-(6\hat{i}-2\hat{j})}{10} = \frac{-3\hat{i}+4\hat{j}}{5}\ \mathrm{m/s^2}$
Force, $F = ma = 5 \times \frac{(-3\hat{i}+4\hat{j})}{5} = (-3\hat{i}+4\hat{j})N$

Q.12 Answer:

187.5

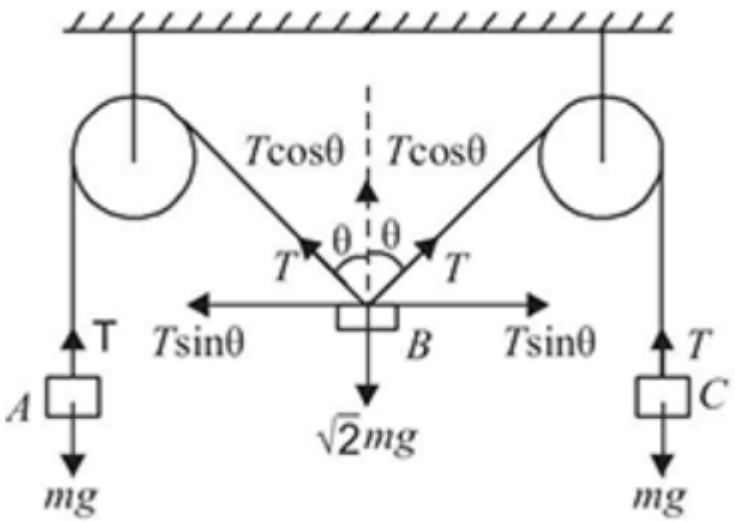
Solution:

Mass of rocket $(\mathrm{m}) = 5000\mathrm{Kg}$ **Exhaust speed** $(\mathrm{v}) = 800\ \mathrm{m/s}$
Acceleration of rocket $= 20\ \mathrm{m/s^2}$ **Gravitational acceleration** $(\mathrm{g}) = 10\ \mathrm{m/s^2}$ **We know that upward force** $\mathrm{F} = \mathrm{m(g+a)} = 5000(10+20) = 5000 \times 30 = 150000\ \mathrm{N}$
We also know that amount of gas ejected $\left(\frac{\mathrm{dm}}{\mathrm{dt}}\right) = \frac{\mathrm{F}}{\mathrm{v}} = \frac{150000}{800} = 187.5\ \mathrm{kg/s}$

Q.13 Answer:

45

Solution:



The tension in both strings will be same due to symmetry. For equilibrium in vertical direction for body B **we have** $\sqrt{2}mg = 2T\cos\theta$
$\therefore \quad \sqrt{2}mg = 2(mg)\cos\theta \quad [\because T = mg$, **(at equilibrium]**
$\therefore \quad \cos\theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$

Q.14 Answer:

$9.2\ \mathrm{kg}$

Solution:

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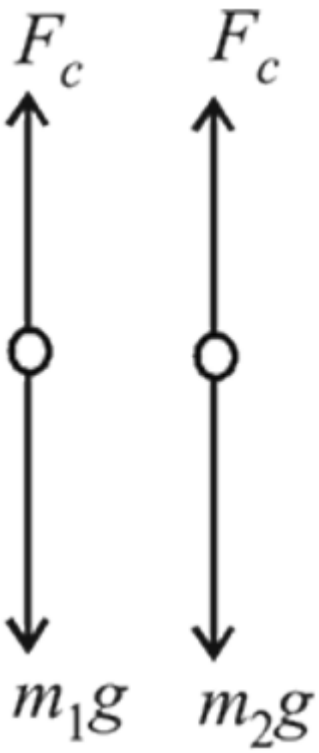
a) Free body diagram at start Free body diagram when $a = 0\ m_1 g - F_c = m_1 a; F_c = m_1(g - a)$

$F_c - m_2 g = 0; F_c = m_2 g$

$m_1(g - a) = m_2 g$

$m_2 = \frac{m_1(g-a)}{g} = 50.8 \text{ kg}$

$\Delta m = 9.2 \text{ kg}$



Q.15 Answer:

0.25

Solution:

Tension at the highest point $\mathrm{T_{top}} = \frac{\mathrm{mv^2}}{\mathrm{r}} - \mathrm{mg} = 2\mathrm{mg} \quad (\therefore \mathrm{v_{top}} = \sqrt{3\mathrm{gr}})$

Tension at the lowest point $\mathrm{T_{bottom}} = 2\mathrm{mg} + 6\mathrm{mg} = 8\mathrm{mg}$

$\therefore \frac{\mathrm{T_{top}}}{\mathrm{T_{bottom}}} = \frac{2\mathrm{mg}}{8\mathrm{mg}} = \frac{1}{4} = 0.25$

Q.16 Answer:

$3 : 1$

Solution:

In case (i) we have

$T_1 - (1 \times g) = 1 \times 4.9$

$\Rightarrow T_1 = 9.8 + 4.9 = 14.7 \text{ N}$

In case (ii), $1 \times g - T_2 = 1 \times 4.9$

$\Rightarrow T_2 = 9.8 - 4.9 = 4.9 \text{ N} \quad \therefore \frac{T_1}{T_2} = \frac{14.7}{4.9} = \frac{3}{1}$

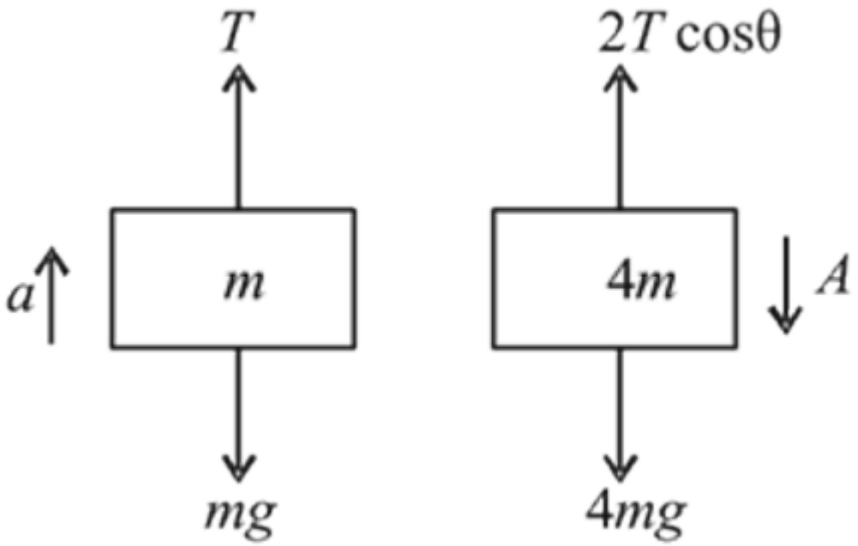
Q.17 Answer:

$\frac{5g}{11}$ downwards

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Solution:

The FBD of blocks is as shown From Newton's second law $4mg - 2T\cos\theta = 4mA$...(i)
and $T - mg = ma$...(ii)
$\cos\theta = 4/5$...(iii)
and from constraint we get $a = A\cos\theta$



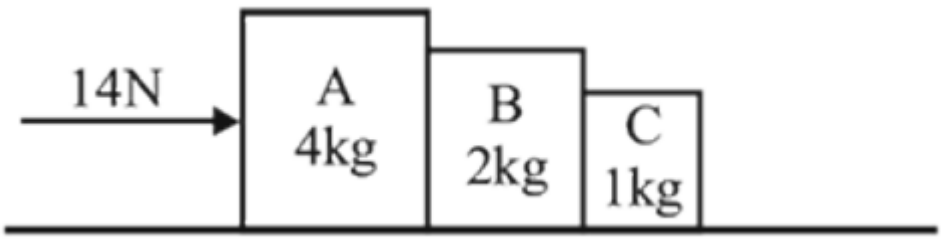
Solving eq. (i), (ii) and (iii) we get acceleration of block of mass $4m$,
$A = \frac{5g}{11}$ **downwards.**

Q.18 Answer:

6 N

Solution:

Acceleration of system $a = \frac{F_{\text{net}}}{M_{\text{total}}} = \frac{14}{4+2+1} = \frac{14}{7} = 2\ \text{m/s}^2$



The contact force between A and $B = (m_B + m_C) \times a = (2+1) \times 2 = 6N$

Q.19 Answer:

$\frac{2ma}{g+a}$

Solution:

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Let upthrust of air be $\mathrm{F_a}$ then For downward motion of balloon $\mathrm{F_a = mg - ma}$

$mg - F_a = ma$

For upward motion $\mathrm{F_a - (m - \Delta m)g = (m - \Delta m)a}$

Therefore $\Delta \mathrm{m} = \frac{2\mathrm{ma}}{g+a}$

Q.20 Answer:

04.90

Solution:

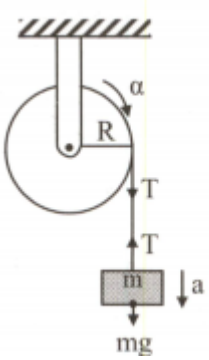
Acceleration, $a = r\alpha \quad \ldots . (i)$

Torque, $\tau = I\alpha \quad \ldots ii$

and $\tau = r \times F = r \times T \quad \ldots \ldots . (\because \mathrm{F = T}) \ldots . . (\mathrm{iii})$**GFrom equation** ii **and** iii**,**

$\mathrm{mr^2}\ \alpha = rT \quad \ldots . 1 = mr^2 \therefore \quad \mathrm{T = mr}\,\alpha$

$T = ma \quad \ldots$ **[From** $i \ldots . . iv$**]**



From the above diagram,

$\mathrm{mg - T = ma} \therefore \quad \mathrm{mg - ma} = ma \quad \ldots .$ **[From** iv**]** $\therefore \quad \mathrm{2ma = mg} \therefore \quad a = \frac{g}{2}$

$= \frac{980}{2} \quad \ldots . . \because \mathrm{g = 9.8m / s^2 = 980cm / s^2}$

$\therefore \quad a = 490\mathrm{cm / s^2}$

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